DIGITAL SAT & EST MATH COURSES

Rabaa Farrag

01120486569

23 YEARS OF EXPERIENCE

Reach a 700+ in Digital SAT & EST Math

✓ FIRST SESSION IS FREE!

Don't watch others hit 700+ while you wait.

EST I MATH • OFFICIAL EXAM • PRACTICE TEST

EST1 2026 Sample

Official Exam • Full Practice Test

Algebra, functions, data analysis, probability and geometry from the official EST I paper. Work through all 50 questions under timed conditions, then check your answers with the answer key and the step-by-step solutions at the end.

50	MCQ	75 min
QUESTIONS	MULTIPLE CHOICE	TIME LIMIT

NAME _____ DATE _____ SCORE _____ /50

How to get the most from this practice test

- **Read** every question slowly and underline the key words.
- **Draw** a quick sketch for every geometry question before you solve it.
- **Solve** step by step and show your work.
- **Check** your answer by substituting it back before you choose.

RF

YOUR MATH COACH

Eng. Rabaa Farrag

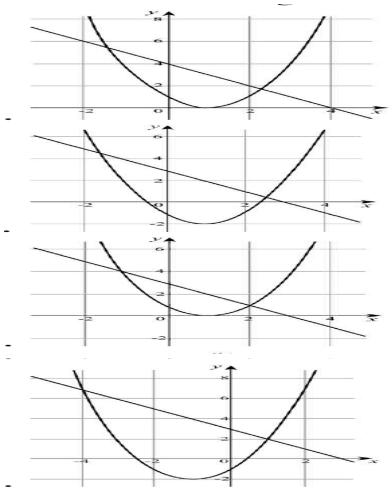
Digital SAT & EST Math expert with **23 years of experience**, helping students master math and reach **700+**.

EST I MATH • OFFICIAL EXAM • 2026 SAMPLE

Practice Questions

50 QUESTIONS

1 Consider the two functions $f(x) = x^2 - 2x + 1$ and $g(x) = -x + 3$. Which of the following is the correct graph representing f and g?



(A) Graph A
(B) Graph B
(C) Graph C
(D) Graph D

2 A box contains 6 plain chocolate bars, 11 chocolates filled with nuts, and 7 chocolates filled with nougat. If Arjun selects one chocolate at random, what is the probability that he chooses a plain chocolate bar?

(A) 0.25
(B) 0.46
(C) 0.55
(D) 0.71

3 A pencil case contains 4 blue pens, 1 green pen, 2 black pens and 2 red pens. One pen is chosen randomly. What is the probability of not picking a blue pen?

(A) 3/5
(B) 5/9
(C) 4/9
(D) 1/3

4 The ratio of men to women in an office is 7 to 4. If there are 66 employees in this office, what is the difference in numbers between the two genders?

(A) 12
(B) 14
(C) 16
(D) 18

Rabaa Farrag

5

$$\frac{2x-7}{5}=\frac{4x^2+7}{10x}$$

5. What is the value of x in the equation above?

A. 1.0
B. −0.5
C. −1.0
D. −2.0

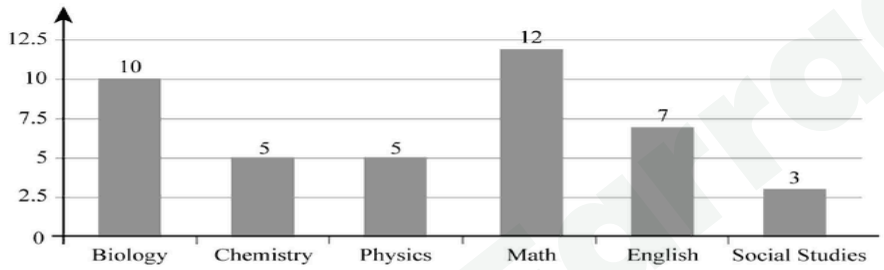
6

Given $p=-1+2i$ and $q=2-3i$, then $p+\bar{q}=$

(A) 1 - 5i
(B) 1 + 4i
(C) 1 + 5i
(D) 5 - i

7

Questions 7 and 8 refer to the information below:



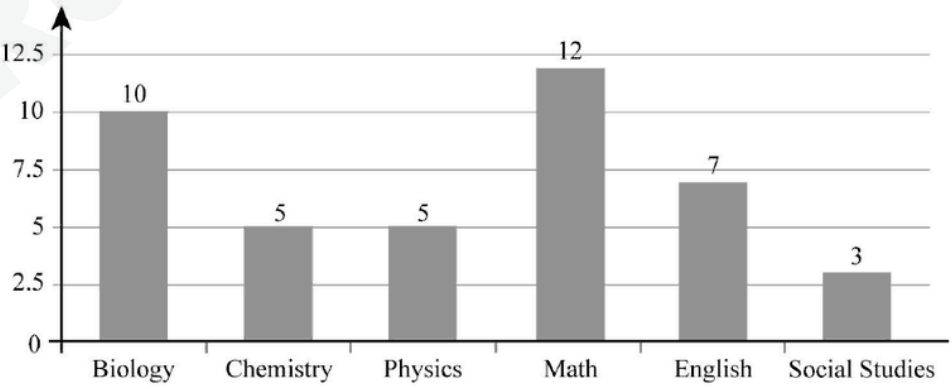
The bar chart above represents the preferred subject chosen by Grade 10 students in a school.

7. What is the approximate percentage of students who prefer to study English?

A. 10.8%
B. 16.7%
C. 28.6%
D. 27.5%

8

Based on the bar chart below, if one student is selected at random, what is the probability that the student prefers either Mathematics or Social Studies?



(A) 5/42
(B) 3/14
(C) 5/14
(D) 5/21

Rabaa Farrag

9 A company manufactures cotton T-shirts. The manufacturing cost for two shirts is $5 and the average worker's wage to produce one shirt is 2. Which of the following equations below best represents the total cost C of producing r shirts?

(A) C = 5r
(B) C = 5r - 2
(C) C = 4.5r
(D) C = 2.5r + 2

10 **10.** Which of the following is the remainder when dividing $3x^4 - 4x^2 + 3x - 6$ by $(x + 2)$?

A. -15
B. 10
C. 15
D. 20

11 If $f(x) = 3x - 5$ and $g(x) = -2x + 1$, which of the following is $f(g(x))$?

(A) -6x - 2
(B) -2x + 5
(C) 4x - 2
(D) 6x - 1

12 **12.** A line passes through the origin and the point $A(2, 28)$. What is the y-coordinate of a point B that lies on the same line if $x_B = -5$?

A. -70
B. -7
C. 0
D. 70

13 The line of equation $y - 1 = 7x$ passes through the point $(1, a)$. What is the value of $2a$?

(A) 8
(B) 10
(C) 16
(D) 18

14 Based on the data in the table (2023: 32 cars, 2024: 40 cars), the owner estimates that the percentage increase in cars sold between 2023 and 2024 would be half the percentage increase between 2024 and 2025. How many cars are expected to be sold in 2025?

Year	Cars sold
2023	32
2024	40

(A) 48
(B) 60
(C) 65
(D) 90

Rabaa Farrag

15 The sum of 5, 15, 2, 18, 7, and x is 50. What is the value of the interquartile range?

A) 12
B) 13
C) 15
D) 16

16 What is the value of x if 25^{rac{3}{x}} = 125^{rac{5}{x} - 4}?

A) 0.50
B) 0.75
C) 1.25
D) 2.50

17 Consider the line l defined by the equation $x - 3y - 12 = 0$. Let line AB be parallel to l, where A has coordinates $(0, -1)$. Which of the following could be the coordinates of point B?

A) (6, -1)
B) (-3, 1)
C) (3, 0)
D) (-2, 1)

18 After a 45% discount, Tim bought sweatpants for $5. What was the original price of the sweatpants before discount?

A) $2.25
B) $7.25
C) $7.50
D) $9.09

19 James has 3 rings, 4 watches, and 2 necklaces. How many different ways can he wear exactly one accessory from each category?

A) 12
B) 20
C) 24
D) 32

20 **20.** Given $f(x) = 2x^2$ and $g(x) = 10x - n$, for which value of n is g tangent to f?

A. -18.5
B. -12.5
C. 12.5
D. 18.0

21 A ball was launched, from the ground, having its trajectory along the curve represented by $y = -3x^2 + 6x + 9$, where x and y are measured in meters. How high, above the ground, does the ball get to as it travels?

A) 12 meters
B) 10 meters
C) 8 meters
D) 5 meters

Rabaa Farrag

22 What is the x-intercept of the line defined by $2x + 5y = 40$?

(A) 2
(B) 8
(C) 10
(D) 20

23 **23.** Given two points $A(-20, 4)$ and $K(12, -3)$, the x-coordinate of the midpoint of $\overline{AK}$ is:

A. -32
B. -4
C. 0.5
D. 16

24 **24.** Which of the following expressions is equivalent to $\frac{x^2+2x+4}{x+5}$?

A. $x - 4 - \frac{18}{x-5}$
B. $x - 3 - \frac{19}{x+5}$
C. $x - 3 + \frac{19}{x+5}$
D. $x + 3 + \frac{19}{x+5}$

25 **Questions 25 – 27 refer to the information below:**

A student went to the bookstore to buy x notebooks and y copybooks for the school. The price of each notebook is \$5 and the price of each copybook is \$6.

25. Which of the following expressions represent the total price of buying y copybooks and x notebooks?

A. $6x + 6y$
B. $5x + 6y$
C. $5x + y$
D. $x + 6y$

26 What is the maximum number of copybooks $(6 each) that can be purchased with 23$?

(A) 10
(B) 4
(C) 3
(D) 1

27 If the student paid $27 at the counter for notebooks (5) and copybooks (\$6), how many of each could he have bought?

(A) 1 notebook and 4 copybooks
(B) 2 notebooks and 4 copybooks
(C) 3 notebooks and 2 copybooks
(D) 3 notebooks and 3 copybooks

Rabaa Farrag

28

$$\left[\frac{(2m^{10}n^{14})^4}{(3m^4n^2)^3}\right]^5$$

28. What is the exponent of n in the simplified form of the expression above?

A. 140
B. 250
C. 300
D. 360

29

Suppose you have a bag of 12 yellow marbles and 9 blue marbles. If you select 2 marbles randomly and without replacement, what is the probability that both marbles are yellow?

(A) 11/35
(B) 16/49
(C) 21/23
(D) 140/157

30

The vertex form of the equation $y = 4x^2 + 12x - 5$ is $y = 4(x + b)^2 - c$. What is the value of $c - b - 3$?

(A) 8.0
(B) 9.5
(C) 10.5
(D) 14.0

31

Amjad and Ali are preparing for a math exam. The probability of Amjad passing is 0.65. The probability of both Ali and Amjad failing is 0.25. Assuming independent events, what is the probability, rounded to the nearest tenth, that both pass?

(A) 0.6
(B) 0.3
(C) 0.2
(D) 0.1

32

A survey was conducted to determine the number of pens owned by employees in a company. The table below shows the number of employees who own 1, 2, 3, and 5 pens. What is the product of the mean, median, and mode of the number of pens?

Number of pens	1	2	3	5
Number of employees	23	17	25	10

(A) 10.25
(B) 13.50
(C) 14.50
(D) 14.56

Rabaa Farrag

33 **33.** $9^2 + 4[6 \times (-5) + 7] + 10 =$

A. -2
B. -1
C. 2
D. 6

34 If the point (a, b), where $a > 1$ and both a and b integers, lies in the solution set of $x + y < 13$ and $y - x > 7$, what is the value of b^a?

(A) 100
(B) 225
(C) 1,000
(D) 1,024

35 A company sells electric scooters for $625 each. Last month, they decided to apply a 10% tax fee on all products. This month, they reduced the total price of a scooter by 15%. Approximately, how much does a scooter cost now?

(A) $531.2
(B) $584.4
(C) $593.7
(D) $687.5

36 To watch a movie, Fady bought 4 adult tickets and 1 child ticket for $46, while Samer bought 3 adult tickets and 2 child tickets for 42$. What is the price of 1 child ticket?

(A) $5
(B) $6
(C) $8
(D) $10

37 In Turkey, the dollar exchange rate increased from 32.2 TRY in May 2024 to 38.5 TRY in May 2025 due to the recent economic crisis. What was the percentage increase in the value of the dollar in TRY, rounded to the nearest whole percent?

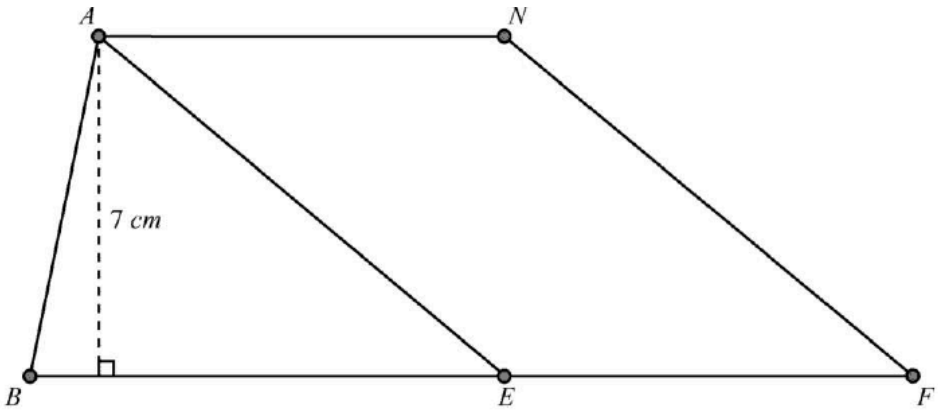
(A) 15%
(B) 19%
(C) 20%
(D) 30%

38 Consider triangle ABC right-angled at A. If $\sin(C) = \frac{\sqrt{2}}{2}$, what is the measure of angle B?

(A) $60°$
(B) $45°$
(C) $30°$
(D) $15°$

Rabaa Farrag

39 In the figure, the area of triangle ABE is $31.5\ \text{cm}^2$. If $FE = \frac{2}{3}BE$, what is the area of the parallelogram ANFE?



(A) 18 cm²
(B) 21 cm²
(C) 42 cm²
(D) 54 cm²

40

$$f(x) = \frac{2}{x+4} - 1$$

40. The vertical asymptote of the graph of f is $x = a$, while its horizontal asymptote is $y = b$. What is the value of $|b + a|$?

A. 1
B. 3
C. 4
D. 5

41

41. If $\frac{13}{8}a + 7x = \frac{1}{2}$ and $x = 1$, what is the value of a?

A. -2
B. -3
C. -4
D. -8

42 The interior angles of a hexagon are in the ratio 2:3:4:5:5:6. What is the measure of the largest angle in this hexagon?

(A) 118.5°
(B) 136.0°
(C) 172.5°
(D) 172.8°

Rabaa Farrag

43

City	Number of staff members
Cairo	33
Alexandria	37
Port Said	19
Giza	30
Aswan	11
Luxor	10

43. A company has 6 different stores in Egypt. The number of staff members in each store is represented in the table above. If the store in Giza decided to increase the number of its staff members to 42, which of the following is true regarding the new average number of staff members?

A. The average will decrease
B. The average will increase by 2
C. The average will increase by 4
D. The average will not change

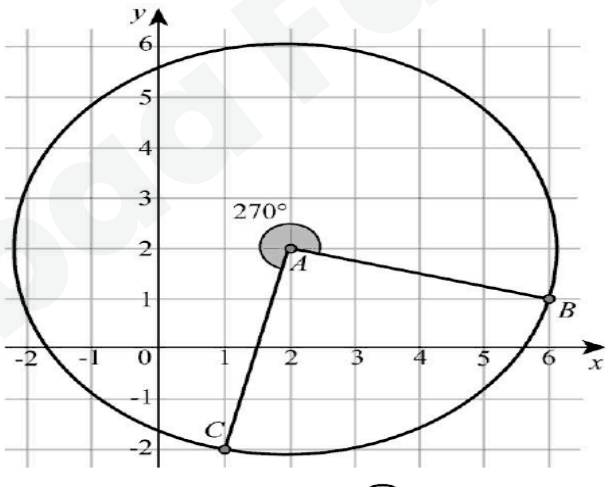
44

44. What is the value of x if $4|2x + 1| = 10$ and $x < 0$?

A. 1.75
B. 0.75
C. -1.75
D. -2.0

45

The area of the sector containing the major arc BC in the figure is:



A \sqrt{17}/4 \pi squared units
B 17/4 \pi squared units
C 51/4 \pi squared units
D 17\pi squared units

Rabaa Farrag

46 Which of the points below does not satisfy the system of inequalities: $2x - y \leq 4$ and $y + 4x > 1$?

(A) (-1, 6)
(B) (1, 2)
(C) (0, 5)
(D) (-1, 1)

47 **47.** What is the value of x if $\frac{3}{x-3} = \sqrt{3}$?

A. $\sqrt{3}$
B. 3
C. $3 + \sqrt{3}$
D. $\sqrt{6}$

48

$$2x^3 + 2x^2 + 4x - 8 = (x^2 + bx + 4)(2x - a)$$

48. In the identity above, true for every real x, a and b are two integers. Which of the following expressions has the smallest value?

A. $-4a - b$
B. $a - 8$
C. $2b - 12$
D. $b^2 - 8$

49 The domain of definition of the function $h(x) = \frac{6}{\sqrt{-2x+7}}$ is:

(A) $(-\infty, 3.5)$
(B) $(0, +\infty)$
(C) $[3.5, +\infty)$
(D) $(3.5, +\infty)$

50

$$4\sqrt{x - 4} = x$$

50. What is the number of real solutions in the equation above?

A. 3
B. 2
C. 1
D. 0

Rabaa Farrag

2026 Sample • CHECK YOUR WORK

Answer Key

50 ANSWERS

Q	ANSWER	Q	ANSWER
1	C	26	C
2	A	27	C
3	B	28	B
4	D	29	A
5	B	30	B
6	C	31	C
7	B	32	D
8	C	33	B
9	C	34	A
10	D	35	B
11	A	36	B
12	A	37	C
13	C	38	B
14	B	39	C
15	A	40	D
16	B	41	C
17	C	42	D
18	D	43	B
19	C	44	C
20	C	45	C
21	A	46	D
22	D	47	C
23	B	48	A
24	C	49	A
25	B	50	C

Rabaa Farrag

2026 SAMPLE • STEP-BY-STEP

Detailed Solutions

50 SOLUTIONS

1 **Answer: C** Graph C

$f(x) = x^2 - 2x + 1 = (x-1)^2$, which is a parabola opening upwards with its vertex at $(1, 0)$. $g(x) = -x + 3$ is a line with a y-intercept of 3 and a slope of -1. Graph C correctly shows this parabola and line.

2 **Answer: A** 0.25

The total number of chocolates is $6 + 11 + 7 = 24$. The number of plain chocolate bars is 6. The probability of choosing a plain chocolate bar is $\frac{6}{24} = \frac{1}{4} = 0.25$.

3 **Answer: B** 5/9

The total number of pens is $4 + 1 + 2 + 2 = 9$. The number of pens that are NOT blue is $1 + 2 + 2 = 5$. The probability of not picking a blue pen is $5/9$.

4 **Answer: D** 18

Let the number of men be $7x$ and the number of women be $4x$. The total number of employees is $7x + 4x = 11x$. We are given $11x = 66$, so $x = 6$. The number of men is $7(6) = 42$ and the number of women is $4(6) = 24$. The difference is $42 - 24 = 18$.

5 **Answer: B**

See the answer key.

6 **Answer: C** 1 + 5i

The conjugate of q is $\bar{q} = 2 + 3i$. Then $p + \bar{q} = (-1 + 2i) + (2 + 3i) = 1 + 5i$.

7 **Answer: B**

See the answer key.

8 **Answer: C** 5/14

Math = 12, Social Studies = 3. Total for both = 15. Probability = $15/42 = 5/14$.

Rabaa Farrag

9 **Answer: C** C = 4.5r

Manufacturing cost per shirt = $\$5/2 = \$2.50$. Wage per shirt = $\$2$. Total cost per shirt = $2.50 + 2 = \$4.50$. Total cost $C = 4.5r$.

10 **Answer: D**

See the answer key.

11 **Answer: A** -6x - 2

$f(g(x)) = 3(-2x + 1) - 5 = -6x + 3 - 5 = -6x - 2$.

12 **Answer: A**

See the answer key.

13 **Answer: C** 16

Substitute $x = 1, y = a$ into the equation: $a - 1 = 7(1) \Rightarrow a = 8$. Then $2a = 16$.

14 **Answer: B** 60

Increase from 2023 to 2024 = $(40 - 32)/32 = 8/32 = 25\%$. This is half the 2024-2025 increase, so the 2024-2025 increase is 50%. Cars in 2025 = $40 \times 1.50 = 60$.

15 **Answer: A** 12

Sum = $47 + x = 50 \Rightarrow x = 3$. The ordered data set is: 2, 3, 5, 7, 15, 18. The lower half is 2, 3, 5 (Q1 = 3). The upper half is 7, 15, 18 (Q3 = 15). IQR = $Q3 - Q1 = 15 - 3 = 12$.

16 **Answer: B** 0.75

Express both sides with base 5: (5^2)^{rac{3}{x}} = (5^3)^{rac{5}{x} - 4}. This simplifies to 5^{rac{6}{x}} = 5^{rac{15}{x} - 12}. Equate the exponents: rac{6}{x} = rac{15}{x} - 12. Multiply by x: $6 = 15 - 12x$. Solve for x: $12x = 9 Rightarrow x = 0.75$.

17 **Answer: C** (3, 0)

Line l: $3y = x - 12 \Rightarrow y = 1/3x - 4$. Slope is $1/3$. Line AB has slope $1/3$ and passes through $(0, -1)$. Equation of AB: $y = 1/3x - 1$. Test the options: For $(3, 0)$, $0 = 1/3(3) - 1 = 1 - 1 = 0$. This point lies on the line.

Rabaa Farrag

18 **Answer: D** $9.09

Let original price be P. $P \times (1 - 0.45) = 5 \Rightarrow 0.55P = 5 \Rightarrow P = 5/0.55 = 500/55 = 100/11 \approx 9.09$.

19 **Answer: C** 24

By the fundamental counting principle, the number of combinations is $3 \times 4 \times 2 = 24$.

20 **Answer: C**

See the answer key.

21 **Answer: A** 12 meters

The maximum height is the y-coordinate of the vertex. $x = -b/(2a) = -6/(2 \times -3) = 1$. Substitute $x = 1$ into the equation: $y = -3(1)^2 + 6(1) + 9 = -3 + 6 + 9 = 12$.

22 **Answer: D** 20

To find the x-intercept, set $y = 0$: $2x + 5(0) = 40 \Rightarrow 2x = 40 \Rightarrow x = 20$.

23 **Answer: B**

See the answer key.

24 **Answer: C**

See the answer key.

25 **Answer: B**

See the answer key.

26 **Answer: C** 3

$23/6 = 3.833...$ The maximum whole number of copybooks is 3.

27 **Answer: C** 3 notebooks and 2 copybooks

We need $5x + 6y = 27$. Let's test the options: A) $5(1) + 6(4) = 29$. B) $5(2) + 6(4) = 34$. C) $5(3) + 6(2) = 15 + 12 = 27$. Option C works.

Rabaa Farrag

28 **Answer: B**

See the answer key.

29 **Answer: A** 11/35

Total marbles = $12 + 9 = 21$. P(first is yellow) = $12/21$. P(second is yellow given first was yellow) = $11/20$. P(both yellow) = $(12/21) \times (11/20) = (4/7) \times (11/20) = 44/140 = 11/35$.

30 **Answer: B** 9.5

Complete the square: $y = 4(x^2 + 3x) - 5 = 4(x^2 + 3x + 2.25) - 5 - 4(2.25) = 4(x + 1.5)^2 - 14$. So $b = 1.5$ and $c = 14$. Then $c - b - 3 = 14 - 1.5 - 3 = 9.5$.

31 **Answer: C** 0.2

P(Amjad fails) = $1 - 0.65 = 0.35$. P(both fail) = P(Amjad fails) $\times$ P(Ali fails) = $0.35\times$ P(Ali fails) = 0.25. P(Ali fails) = $0.25/0.35 = 5/7$. P(Ali passes) = $1 - 5/7 = 2/7$. P(both pass) = $0.65 \times (2/7) = 1.3/7 \approx 0.1857 \approx 0.2$.

32 **Answer: D** 14.56

Total employees = $23 + 17 + 25 + 10 = 75$. Sum of pens = $1(23) + 2(17) + 3(25) + 5(10) = 23 + 34 + 75 + 50 = 182$. Mean = $182/75$. Median is the 38th value. First 23 are '1', next 17 are '2' (up to 40th). So median = 2. Mode is 3 (highest frequency 25). Product = $(182/75) \times 2 \times 3 = 182/75 \times 6 = 1092/75 = 14.56$.

33 **Answer: B**

See the answer key.

34 **Answer: A** 100

We have $y < 13 - x$ and $y > x + 7$. So $x + 7 < y < 13 - x$. This requires $x + 7 < 13 - x \Rightarrow 2x < 6 \Rightarrow x < 3$. Since $x > 1$ and is an integer, $x = 2$. Then $2 + 7 < y < 13 - 2 \Rightarrow 9 < y < 11$. Since y is an integer, $y = 10$. So $a = 2, b = 10$. $b^a = 10^2 = 100$.

35 **Answer: B** $584.4

Price after tax = $625 \times 1.10 = 687.5$. Price after reduction = $687.5 \times (1 - 0.15) = 687.5 \times 0.85 = 584.375 \approx \$584.4$.

36 **Answer: B** $6

System: $4A + C = 46$ and $3A + 2C = 42$. From first, $C = 46 - 4A$. Substitute into second: $3A + 2(46 - 4A) = 42 \Rightarrow 3A + 92 - 8A = 42 \Rightarrow -5A = -50 \Rightarrow A = 10$. Then $C = 46 - 4(10) = 6$.

Rabaa Farrag

37 **Answer: C** 20%

The percentage increase is $\frac{\text{New}-\text{Old}}{\text{Old}} \times 100\% = \frac{38.5-32.2}{32.2} \times 100\% = \frac{6.3}{32.2} \times 100\% \approx 0.1956 \times 100\% = 19.56\%$. Rounded to the nearest whole percent, this is 20%.

38 **Answer: B** 45°

Since $\sin(C) = \frac{\sqrt{2}}{2}$, angle C is 45°. In a right triangle, the sum of the two acute angles is 90°. Therefore, angle $B = 90^\circ - 45^\circ = 45^\circ$.

39 **Answer: C** 42 cm²

Area of $\triangle ABE = \frac{1}{2} \times BE \times h = 31.5$. Area of parallelogram $ANFE = \text{base} \times \text{height} = FE \times h$. Since $FE = \frac{2}{3}BE$, Area $= \frac{2}{3}BE \times h = \frac{4}{3} \times (\frac{1}{2} \times BE \times h) = \frac{4}{3} \times 31.5 = 42\ \text{cm}^2$.

40 **Answer: D**

See the answer key.

41 **Answer: C**

See the answer key.

42 **Answer: D** 172.8°

Sum of interior angles of a hexagon = $(6-2) \times 180^\circ = 720^\circ$. Sum of ratio parts = $2+3+4+5+5+6 = 25$. Largest angle = $(6/25) \times 720^\circ = 172.8^\circ$.

43 **Answer: B**

See the answer key.

44 **Answer: C**

See the answer key.

45 **Answer: C** 51/4 \pi squared units

Radius $r^2 = (2-1)^2 + (2-(-2))^2 = 1^2 + 4^2 = 17$. Area of circle = 17π. The major arc is 270°, which is $270/360 = 3/4$ of the circle. Area of sector = $(3/4) \times 17\pi = 51\pi/4$.

Rabaa Farrag

46 **Answer: D** (-1, 1)

Test (-1, 1): $2(-1) - 1 = -3 \leq 4$ (True). $1 + 4(-1) = -3 > 1$ (False). So (-1, 1) does not satisfy the system.

47 **Answer: C**

See the answer key.

48 **Answer: A**

See the answer key.

49 **Answer: A** $(-\infty, 3.5)$

For the function to be defined, the expression inside the square root must be strictly greater than zero (since it's in the denominator). So, $-2x + 7 > 0$. Solve for x: $-2x > -7 \Rightarrow x < 3.5$. In interval notation, this is $(-\infty, 3.5)$.

50 **Answer: C**

See the answer key.

Rabaa Farrag